

Name: Vishesh Kansugia

UID: 24BAI70184

Subject: ADBMS

Question 1:

Write an SQL solution to calculate the total number of bank accounts belonging to each salary category. The classification categories are:

1. **"Low Salary"**: All monthly salaries strictly less than \$20,000.
2. **"Average Salary"**: All monthly salaries in the inclusive range [\$20,000, \$50,000].
3. **"High Salary"**: All monthly salaries strictly greater than \$50,000. The final result table must report all three categories, even if a category contains zero accounts.

Answer :

```
12
13 select 'Low Salary' as category, count(account_id) accounts_count
14 from Accounts
15 where income<=20000
16 UNION ALL
17 select 'Average Salary' as category, count(account_id) accounts_count
18 from Accounts
19 where income>20000 and income<=50000
20 UNION ALL
21 select 'High Salary' as category, count(account_id) accounts_count
22 from Accounts
23 where income>50000;
24
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	category text	accounts_count bigint
1	Low Salary	1
2	Average Sala...	1
3	High Salary	3

Total rows: 3 Query complete 00:00:06.384 LF Ln 13, Col 64

Question 2:

Design an Employee Management System database architecture by implementing the following operations sequentially:

1. Create and populate structural tracking tables for Departments, Employees, and Salaries.
2. Query the system using multi-table Joins to pull clear department breakdowns, apply a 10% salary enhancement across all members of the HR department, and delete any individual salary record dropping strictly below \$30,000.
3. List all remaining active employees whose personal salary exceeds the overall company-wide average salary.
4. Part B: Apply 10% HR increase using IN subquery filtering

Answer :

```
28 select * from Employees e
29 left join Salaries s on e.emp_id=s.emp_id
30 left join Departments d on e.dept_id=d.dept_id;
```

	emp_id integer	name character varying (50)	dept_id integer	emp_id integer	salary integer	dept_id integer	dept_name character varying (50)
1	1	Alice	10	1	50000	10	HR
2	2	Bob	10	2	25000	10	HR
3	3	Charlie	20	3	80000	20	IT
4	4	David	20	4	40000	20	IT
5	5	Emma	30	5	45000	30	Sales

Total rows: 5 Query complete 00:00:01.498

```
32 -- 2
33 update Salaries set salary=salary+salary*0.1 where emp_id
34 IN (Select emp_id from Employees e join Departments d
35 on e.dept_id=d.dept_id where dept_name='HR');
36
37 DELETE from Salaries where salary<30000;
38 Select * from Salaries;
39
40 -- 3.
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	emp_id [PK] integer	salary integer
1	3	80000
2	4	40000
3	5	45000
4	1	55000

Total rows: 4 Query complete 00:00:03.953 LF Ln 37, Col 41

Question 3:

A pizza restaurant stores all the available pizza toppings and their ingredient costs in a table called `pizza_toppings`.

Every customer must choose **exactly 3 different toppings** to prepare a pizza.

Write a PostgreSQL query to generate **every possible 3-topping pizza combination** along with its **total ingredient cost**.

Requirements

Every pizza must contain **exactly 3 different toppings**.

The same topping **cannot appear more than once** in a pizza.

Different arrangements of the same toppings should be considered the **same pizza**.

Answer:

```

14 select CONCAT(T1.topping_name,',',T2.topping_name,',',T3.topping_name) as pizza,
15 (T1.ingredient_cost + T2.ingredient_cost+ T3.ingredient_cost) as total_cost from pizza_toppings T1
16 join pizza_toppings T2
17 on T2.topping_name>T1.topping_name
18 join pizza_toppings T3
19 on T3.topping_name>T2.topping_name
20 Order by total_cost DESC
21

```

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1 of 1

	pizza text	total_cost numeric
1	Chicken,Pepperoni,Sausage	1.75
2	Chicken,Extra Cheese,Sausage	1.65
3	Extra Cheese,Pepperoni,Saus...	1.60
4	Chicken,Onions,Sausage	1.50
5	Chicken,Extra Cheese,Pepper...	1.45
6	Onions,Pepperoni,Sausage	1.45
7	Extra Cheese,Onions,Sausage	1.35
8	Chicken,Onions,Pepperoni	1.30
9	Chicken,Extra Cheese,Onions	1.20

Total rows: 10 Query complete 00:00:05.974